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[Project Title -'Campus Compass AI' : Smart Institutional Navigation Bot]

Project Domain - Institutional Knowledge and Policy Management

[e.g., Excel, Microsoft Copilot Studio with Copilot]

A Project Report

Submitted in partial fulfilment of the requirements of the AICW Project

by:

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Firstly I would like to thank god for choosing me for this opportunity to be a part of this prestigious project.

Next, would like to express my deep sense of gratitude to all stakeholders –mentors, teachers, guides, and fellows etc who helped me directly or indirectly during the project work – ‘Campus Compass AI - : Smart Institutional Navigation Bot’.

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Finally, I am thankful to my family and friends for their continuous encouragement and support throughout the tenure of the project.

ABSTRACT OF THE PROJECT

This project is part of the **AICW (Artificial Intelligence in the Contemporary World)** Fellowship Program, a collaborative initiative by **Edunet Foundation** in partnership with **Microsoft, LinkedIn**, and **SAP**. The fellowship is designed to equip students with practical, hands-on skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

My project entitled – '**Campus Compass AI - Smart Institutional Navigation Bot'.** - focuses on identifying the challenges and problems faced by faculty members while navigating their way in higher education institutions. In the age of automation and AI, most of the HEI's are dependent on lengthy paperwork, overwhelming workload, conflicting responsibilities due to class schedule, administrative duties, student queries and so on thereby reducing their output and motivation. The following **tools** have been used:-

- **Survey Questionnaire** → Circulated to 201 faculty members across the university to capture real experiences and pain points.
- **Excel** → Data collation, cleaning, and preparation for analysis.
- **Interactive Dashboards (Excel)** → Visualized faculty challenges, satisfaction scores, and workload metrics.
- **Microsoft Copilot** → Assisted in deriving insights and building dashboards.
- **Microsoft Copilot Studio** → Developed a chatbot capable of addressing common faculty queries in real time.

The project undertaken focused on making their life easy at two levels. I have firstly identified the problems through a survey questionnaire circulated to 201 faculty members, then collated the data through excel , after cleaning the data and interactive dashboards I have identified the common challenges faced by teachers as well as some recommendations. The development of dashboards was assisted through insights derived from Microsoft Copilot. Secondly I have also developed a chatbot using Microsoft Copilot Studio which can communicate with teachers and solve the common queries in real time.

Hence I can conclude that by utilizing AI assisted chatbot, most of the grey areas while induction can be identified and rectified, problems like identifying department buildings, holidays, academic calendars and specific faculty identification as well as integrating the data with LMS can increase the faculty members' motivation as well as retention and improve work productivity.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement-

The strength, reputation, and ultimate success of any educational institution are fundamentally anchored in the quality of its faculty. Ironically, most of the higher education institutions often face challenges in on boarding and inducting new faculty members, such as **information overload, inconsistent delivery of policies, overlapping events happening simultaneously, delays in access to resources, and low engagement during orientation**. Many of these problems also arise due to nature and scope of employment, namely – **Fixed term tenure, Adjunct or Visiting faculty** and a minority of faculties are also hired on **online or remote basis**. In such circumstances, manual processes like drafting leave applications, emails, PDFs, holiday and academic calendars lead to confusion, higher administrative workload, and suboptimal integration, resulting in reduced faculty productivity and retention.

The inability to efficiently navigate a campus has cascading effects on time management, emotional well-being, and institutional perception. Some of the problems are as follows:-

1. Navigation Constraint– In the current infrastructure of the university, several department buildings are scattered and housing different departments making it more challenging to navigate for newcomers.

2. Workload Balance– Most of the faculties feel the burden of workload as they have to assist in classroom preparation, lectures, academic activities, FDP's , NAAC data compilation and other administrative duties.

3. Data Silos– Existing process includes data concentrated to the hands of a few like– Leave Policy, Holiday Schedule , Exam Schedule , Annual Calendar , Placement Data , Semester Break etc.

4. WorkStress– Due to above pointers it leads to work stress, which impacts productivity and finally leads to attrition.

5. GPS tools Limitation- GPS tools (e.g., Google Maps) are ineffective indoors or for hyperlocal campus navigation

6. Static Websites- Institutional websites and directories are general, static, and not mobile-friendly.

7. Lack of conversational interfaces (NLP)– This is a major hindrance which makes existing tools less engaging and harder to use.

These problem are significant as they directly affect the quality of faculty engagement, awareness of academic activities, available resources and indirectly hampers their performance, motivation to work and retention. The results of the same maybe lower attachment to the institution, and increased attrition rate.

These problems create a ripple effect where all the stakeholders in the academic ecosystem are impacted. The results of the same maybe lower attachment to the institution, and increased attrition rate.

Those who are directly affected by this confusion are the regular and visiting faculties and non-teaching staff as well as students and parents.

The existing tools available offer broad range navigation like google maps which does not work within an institution. Besides websites and web directories offer searchable maps which are not mobile friendly and are quite general lacking basic conversational tone. (NLP)

This project addresses these challenges by applying data cleaning, analysis, visualization, and AI integration techniques using industry-standard tools. The project will demonstrate how structured data workflows combined with AI assistance can produce accurate, actionable insights and outputs relevant to a chosen real-world domain.

1.2 Motivation

a. Personal/Professional Importance - India has approximately **16 lakh** teachers in HEIs, with the majority **(68%) being Lecturers/Assistant Professors** (https://www.education.gov.in/sites/upload_files/mhrd/files/PIB2102789.pdf)

As an academician this topic is of special relevance to me since I faced the puzzling situation of navigating the classrooms as a new appointee. The GPS is not acquainted to a hyperlocal geography like inside a vast 25 acre university. Building 'Campus Compass AI' allows me to apply advanced concepts like Natural Language Processing (NLP) and make this a tangible product able to be accessed by mobile phones.

b. Applications in Real World - Apart from an educational institute this concept can be scaled to larger infrastructure requirements like –

 Healthcare Facilities: Large hospitals like Apollo Hospitals, AIIMS (Delhi), Tata Medical Hospital (TMH) can implement this helping highly stressed patients and visitors find specific wards, pharmacies, or doctor's offices in massive, multi-wing hospital complexes without wandering into restricted areas.
 Corporate Mega campuses/ IT Parks: Electronic City, Bangalore , **HITEC City, Hyderabad** along with corporate giants like **Amazon** and **Google** can use this technology and help new hires, clients, and interviewees in navigating large tech parks, finding specific conference rooms, or locating the nearest cafeteria.
c. Impact on Organization -Implementing a smart navigation bot creates a ripple effect of benefits for the institution such as:

- **Increased Operational Efficiency**
- **Data-Driven Facility Management**
- **Disabled needs support (Wheelchair and Ramp Access)**

d. Alignment with AI and data analytics- This technology uses NLP to answer questions ins a friendly conversational tone setting it apart with GPS based maps. My project focuses on applying **indoor positioning systems (IPS) which goes further beyond (GPS)**

1.2 Objectives

Through my project **,"Faculty Onboarding and Orientation Bot"—an AI-powered chatbot**, I aim to address and correct the following challenges and streamline the process making it easier for new faculties as well as the administration to access data and manage problems real time at the same time delivering valuable experiences for the new faculty members.

A. **Streamline Onboarding Process** – The faculty members can enhance user engagement by applying to leaves, checking upcoming holidays and academic calendar online. They can also check the institution vision, mission, academic activites and goals.

B. **Reduce Administrative Burden:** With the help of an AI based bot which can be accessed through mobile phones hence portable it can offer immediate access to teaching tools, learning **management systems (LMS),** and research resources to help faculty start their roles faster. Staffs can access the following :-

- ***Leave Balance***
- ***Academic Calendars (Cultural Events)***
- ***Annual Holiday List***
- ***Semester Breaks***
- ***Hackathons***
- ***Placement Updates***
- ***College/University Headlines***
- ***Update of Different groups / committees***

C. **Support Career Development & Training –** The chatbot can share information on FDP/MDP programs, technical training, and AI workshops. Every faculty is encouraged to participate in a skill-building opportunities of their choice.

D. **Grievance Redressal -** The bot provides a platform for direct and anonymous grievance redressal for addressing problems and challenges to a specific committee – Grievance Committee and get result in shorter Turnaround time. Their identity will not be disclosed for privacy concerns and their issues will be dealt with in a systematic manner and documented for future references.

E. **Boost Motivation, Retention & Productivity** – This AI agent Improves faculty satisfaction by addressing pain points quickly, it also creates a supportive environment that enhances retention, removing information deficit and overall productivity.

1.4 Scope of the Project

This project entitled – '**Campus Compass AI - Smart Institutional Navigation Bot'**- is conceptualized in two phases. Firstly, a simulated dataset is curated by combining primary dataset derived through a questionnaire circulated to faculty members as well as with co-pilot sourced data to maintain heterogeneity and It includes questions like – Duration in organization, Workload Balance perception, FDP/MDP participation rate, Stress perception index and so on. This has helped in identify various challenges faced by faculties, their problem areas and also recommendations suggested by most of them. Then these challenges and pain points are cumulated and a knowledge base is imported to an agentic AI, thereby creating an AI powered chatbot which aims to address the real-time issues and provide solutions reducing the time gap and offering customized solution to contextual problems.

The tools used are mostly Excel for tabulation and visualization in the form of pivots and charts.

I have taken insights from Microsoft Co-pilot and Google Gemini. Finally the responses are collected and cumulated together to create an AI Chabot.

The project utilizes real-time data integration, predictive machine learning models and GPT 4 to analyse the data and come at a valid conclusion. The project showcases that AI assisted tools when instructed and inculcated with proper data can seamlessly ease the life of faculty members thereby allowing them to focus on better quality teaching.

CHAPTER 2
Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

Chatbots are intelligent conversational computer programs that mimic human conversation in its natural form (Jia, J. The Study of the Application of a Keywords-based Chatbot System on the Teaching of Foreign Languages. *arXiv* **2003**, arXiv:cs/0310018. [**Google Scholar**]) Chatbots can also be defined as "online human-computer dialogue system(s) with natural language" (Cahn, J. CHATBOT: Architecture, Design, & Development. Ph.D. Thesis, University of Pennsylvania, School of Engineering and Applied Science, Philadelphia, PA, USA, 2017. [**Google Scholar**]). Chatbots are currently applied to a variety of different fields and applications, spanning from education to e-commerce, encompassing healthcare and entertainment. Chatbots appear, in fact, to be more engaging to the user than the static Frequently Asked Questions (FAQ) page of a website. At the same time, chatbots can simultaneously assist multiple users, thus resulting more productive and less expensive compared to human customer supports services.

Alan Turing is thought to be the first person to have ***conceptualised the idea of a chatbot in 1950***, when he proposed the question: "Can machines think?". Turing's description of the behaviour of an intelligent machine evokes the commonly understood concept of a chatbot.(Turing, A.M. Computing Machinery and Intelligence. *Mind* **1950**, *LIX*, 433–460. [**Google Scholar**]) Chatbots have evolved with the progressive increase in computational capabilities and advances in **Natural Language Processing** tools and techniques. The first implementation of a chatbot, which relied heavily on linguistic rules and pattern matching techniques, was achieved in **1966** with the development of **ELIZA.** It could communicate with the user through keyword matching program. A marked evolution in chatbot in the **1980s** is the use of Artificial Intelligent. **A.L.I.C.E. (Artificial Intelligent Internet Computer Entity)** is based on the Artificial Intelligence Mark-up Language (AIML), which is an extension of XML.

The application of chatbots has expanded with the emergence of Deep Learning algorithms. One of the new, and the most interesting application, is the development of smart personal assistants (such as Amazon's Alexa, Apple's Siri, Google's Google Assistant, Microsoft's Cortana, and IBM's Watson). Personal assistants chatbots or conversational agents that can usually communicate with the user through voice are usually integrated in smartphones, smartwatches, dedicated home speakers and monitors, and even cars. An interesting example of a social chatbot is Microsoft's XiaoIce. XiaoIce is meant to be a long-term companion to the user, and in order to achieve high user engagement it has been designed to have a personality, an Intelligent Quotient (IQ) and an Emotional Quotient (EQ). Knowledge and memory modelling, image and natural language comprehension, reasoning, generation, and prediction are all examples of IQ capabilities.
There are at least two major limitations that have yet to be addressed. Firstly, there is no common framework for chatbot's evaluation. Although some metrics are widely used for measuring chatbots performance there is no specific metric or set of metrics commonly referred to as the reference. The lack of common frame of reference concerning chatbots' evaluation limits the correct testing and comparison of different models. A vital aspect of chatbots is their application in different industries and to real-life scenarios. Even though some papers provide an overview of chatbots' application and some numbers regarding their increase in popularity on the web, there appears to be a lack of clarity in identifying industry or task-oriented chatbots' applications and

social or companion chatbots. (A Literature Survey of Recent Advances in Chatbots by Guendalina Caldarini Kenneth McGarry)

2.2 Tools and Technologies Review

Microsoft Excel is an extensively used tool in this project for data cleaning, transformation, and basic visualization. Pivot tables, conditional formatting, and data validation are commonly used techniques. Microsoft Excel is used in both academic and professional domains as it is most widely available, easy to use and understand. However, excel alone will not be enough. It needs to be used in combination of several tools which can interpret and visualize data.

Microsoft Copilot, Google Gemini, Chatgpt etc are AI-powered conversational assistant integrated across Microsoft 365 apps and Windows, designed to boost productivity by summarizing information, drafting content, and analyzing data. It enhances research by conducting deep searches, synthesizing documents, and automating complex data analysis tasks in Excel and Power BI. It is a graphical platform used to create, customize, and manage AI agents (previously chatbots) that automate workflows, answer questions, and act on behalf of users

AI chatbots are software applications using natural language processing (NLP), machine learning (ML), and large language models (LLMs) to simulate human conversation and answer queries. They assist in research and analysis by quickly synthesizing information, summarizing large documents, brainstorming ideas, cleaning data, and identifying trends, significantly accelerating workflow efficiency. The best part is anyone without a proper knowledge of coding can also run these apps, that's where AICW along with Edunet, Microsoft and LinkedIn roped in many female faculties who are from Non- STEM background.

2.3 Gaps Identified

Although AI based tools like – AI Agents, Microsoft Co-pilot and Power BI are indispensable in making today's modern workplaces seamless but they have their own pitfalls.

Firstly, The large models that represent the latest improvement in the field, whether they concern language models in general or dialogue models, are not yet suitable to be easily applied by the industry, as they require great computational power and exceptionally large training datasets.

Secondly, although many survey papers on chatbots present a detailed explanation of chatbots' technologies and implementations, recent surveys lack information on most recent advances in language models that might be applied to chatbots, such as Transformers, which we have provided an overview of these advanced module in this paper. Most existing literature focuses on complex machine learning approaches; simpler, tool-based solutions for non-technical users are underexplored.

Finally, a vital aspect of chatbots is their application in different industries and to real-life scenarios. Even though some papers provide an overview of chatbots' application and some numbers regarding their increase in popularity on the web, there appears to be a lack of clarity in identifying industry or task-oriented chatbots' applications and social or companion chatbots.

Consequently, my chatbot is here to stay, and it will answer a few question like –

- Eradicate the ambiguity and decision less ness by providing all the news at the user's fingertips.
- To reduce paperwork and bureaucratic structures existing in the HEI's and make actions more automated.
- Quickly apply for leaves and check upcoming holidays and FDP's thereby allowing the faculty's work-life balance and improve attrition and morale.

CHAPTER 3
Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

My dataset for the project is a combined to faculty data as well as artificially simulated data resulting in 201 faculty responses. These are collecting through input mechanisms like google forms ad them it is cleaned and record captured in Microsoft Excel. The parameters for faculty members include – Employment duration, teaching hours, workload balance index, stress index, FDP/MDP availability, and research grants.

This data is further cleaned by removing null values, overlapping data, unwanted and false entries and then after applying sort and filter function and various formulas creating a summary table which finally leads to the formation of a dashboard. Dashboard is made after adding charts, pivot tables, slicers.

Finally the output is fed to Microsoft co-pilot and interesting suggestions are generated. Then on the basis of recommendations and obstacles, the 'Knowledge Base' is created and fed to the AI chatbots to create an institutional chatbot known for tis friendly and quick response to all the queries a probationary / new faculty ay have.

3.2 Modules / Components Used

List the key modules or functional components of your project:

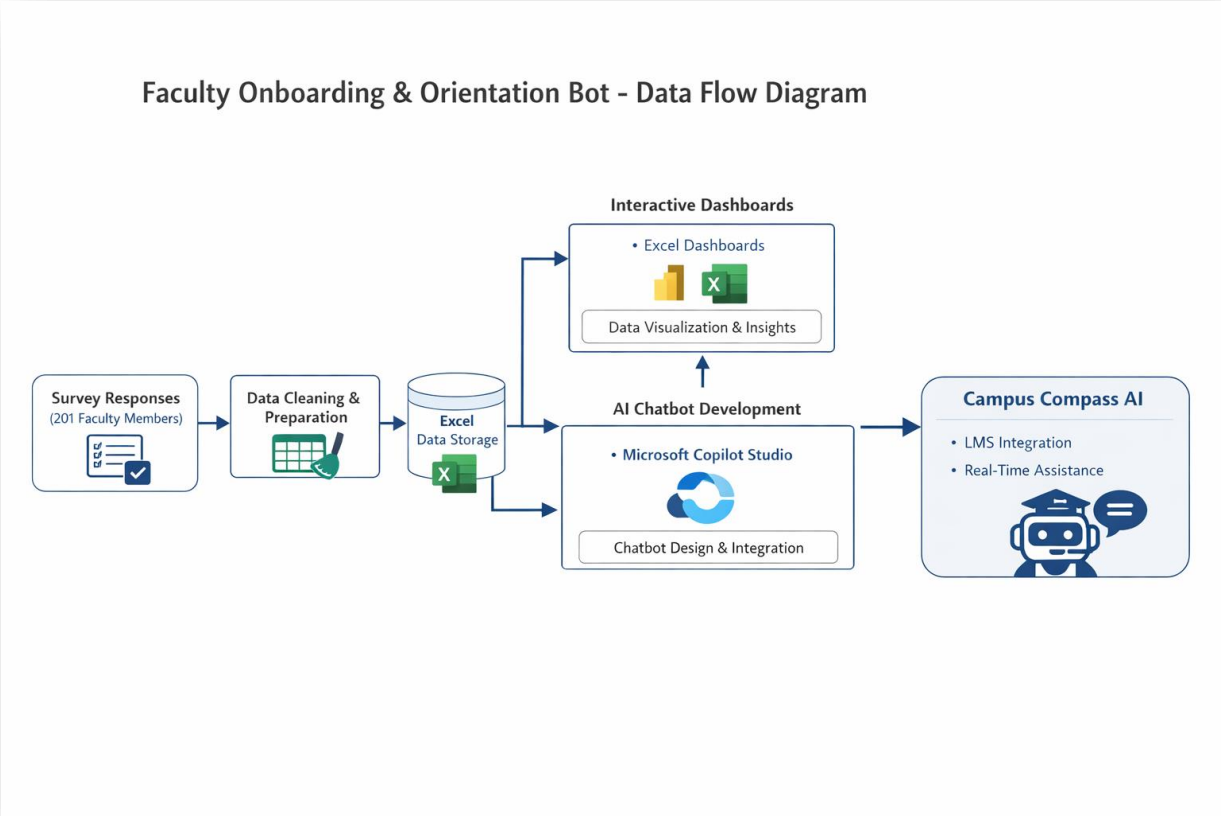
- **Module 1 — Data Ingestion:** A raw dataset is created through the responses of faculty members teaching in various domains and with varied years of experience. This is ten collected and fed into Excel.
- **Module 2 — Data Cleaning:** The excel dataset is then checked for the following errors - inconsistencies, duplicates, null values, false entries etc. Then table is inserted, the data is sorted, filtered and tabulated.
- **Module 3 — Data Analysis:** The cleaned data is then analysed and calculations are done, formulas are applied to come to the final summary sheet.
- **Module 4 — Data Visualization :** Finally with the help of Excel itself and applying simple formulas a visual dashboards is created using pivot tables, charts along with slicers.
- **Module 5** — **AI Integration:** This dashboard generated is then input into Microsoft Copilot and Google Gemini to create insightful summary and recommendations.
- **Module 6** – **Creation of Chatbot** : The dashboards along with the AI assisted responses and suggestions is taken into consideration and a Knowledge base is created , finally all these are compiled and input and a conversational bot is created to solve the real time issues addressing faculty members.

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram

Shows the system as a single process with its inputs and outputs.



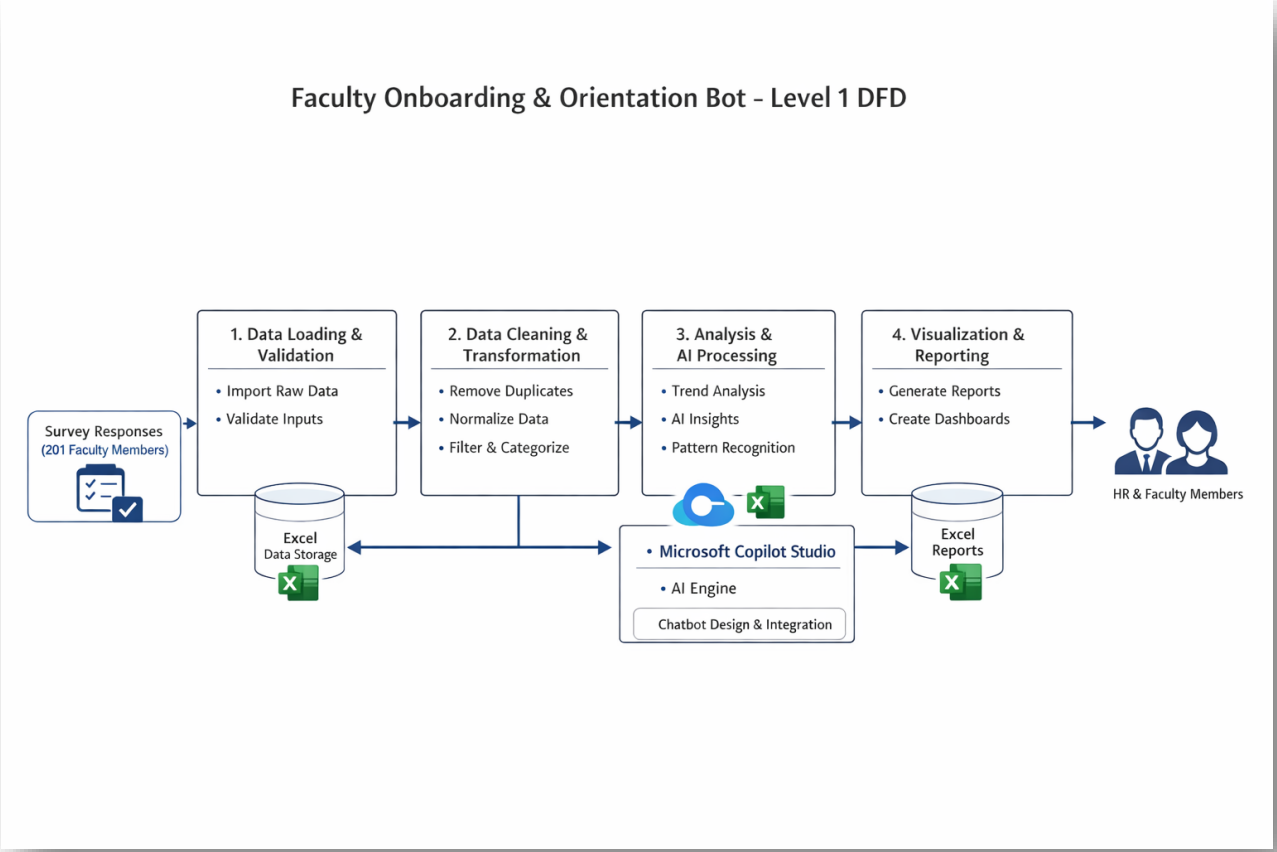
Generated through co-pilot

The schematic description of the above diagram is as follows:

1. **Survey Responses (201 Faculty Members)** → Firstly the responses from the faculty members are collected through data collection devices like google forms and created into a consolidated excel sheet.
2. **Data Cleaning & Preparation (Excel)** → Next, the data is checked for errors like false entries, null values, inconsistencies, formatting errors etc and then cleaned.
3. **Excel Data Storage** →This consolidated cleaned excel sheet forms the base of our project it is saved in a repository.
4. **Interactive Dashboards (Excel)** → Next the excel sheet parameters are grouped into separate heads and then pivot tables and charts are made and exported to a visual summary also known as a Dashboard . This includes tables, KPI cards, summary and slicers.
5. **AI Chatbot Development (Microsoft Copilot Studio)** → Finally the dashboard and AI generated summary is taken and fed into an agent leading to the creation of a AI generated chatbot.

6. **Campus Compass AI** → This leads to the generation of – **'Campus Compass AI – A Smart Navigation Bot**' This is then integrated to LMS to generate *Real-Time Assistance.*

DFD Level 1 — Detailed Process Flow



Generated through co-pilot

Sub processes include the following:

- **Sub-process 1: Data Loading and Validation**

  Excel: Imports raw survey data from faculty responses and validates entries for completeness and format consistency.
  Copilot Studio: Automates data intake by prompting users or administrators to confirm missing or inconsistent inputs before processing

- **Sub-process 2: Data Cleaning and Transformation**

  Excel: Removes duplicates, normalizes categorical values, and structures data for analysis using formulas and filters.

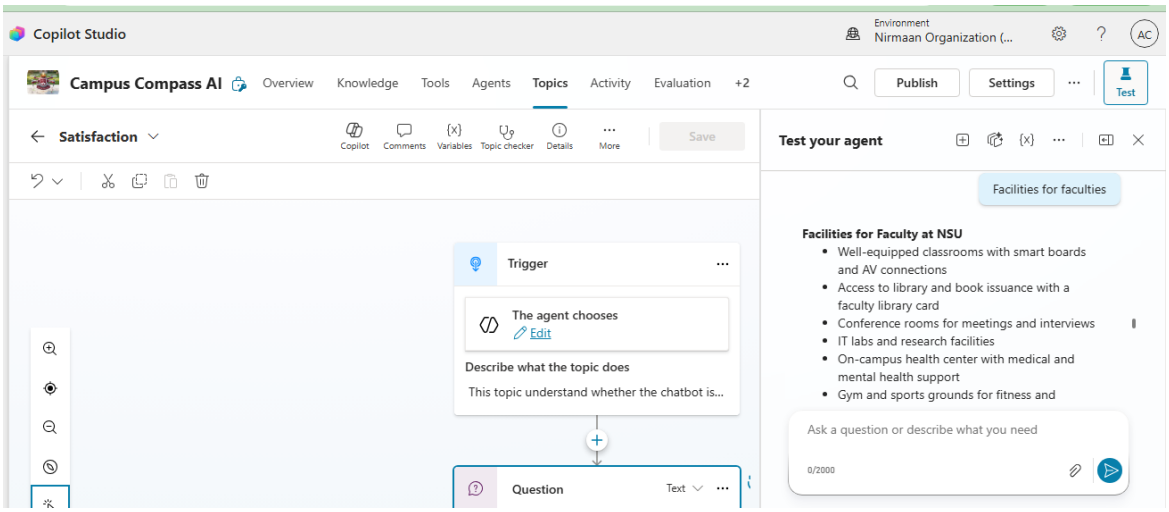
 Copilot Studio: Applies AI-driven logic to categorize responses and flag anomalies, ensuring clean data flows into dashboards.

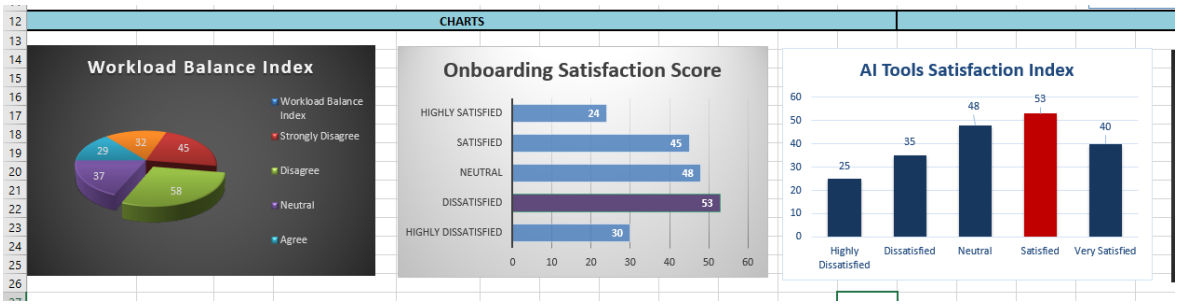
- **Sub-process 3: Analysis and AI Processing**

 Excel: Performs trend analysis and KPI calculations to identify patterns in faculty feedback.
 Copilot Studio: Uses AI reasoning to interpret insights, detect recurring challenges, and feed them into the chatbot's knowledge base.

- **Sub-process 4: Visualization and Report Generation**

 Excel: Generates interactive dashboards and visual reports summarizing satisfaction, workload, and retention metrics.
 Copilot Studio: Integrates these insights into the **Campus Compass AI** chatbot, enabling real-time query responses and data-driven recommendations.





3.4 Advantages

The proposed methodology has several advantages over traditional workload management.

- Automates repetitive data cleaning tasks, saving significant manual effort.
- Provides clear, visual insights that are easy to understand for non-technical stakeholders.
- Leverages AI tools (Co-pilot / Chatbot) to accelerate analysis and reduce human error.

- Scalable — the same methodology can be applied to larger datasets or different domains.(as already explained previously)
- Cost-effective — uses widely available tools (Excel) that are part of the Microsoft 365 ecosystem.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3
RAM	Minimum 4 GB (8 GB)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Enabled (for Copilot / Chatbot access)

Software Requirements

Software	Details
Operating System	Windows 10 pro, version 22 H2
Microsoft Excel	(Microsoft 365 / Excel 2019)
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Through Microsoft 365 subscription (Nirmaan id)
AI Chatbot	Microsoft Copilot (web) / Google Gemini
Web Browser	Microsoft Edge (latest version)

CHAPTER 4
Implementation and Results

This chapter explains how 201 faculty dataset is collected, curated, cleaned and finally tabulated into meaningful charts, pivot tables leading to a visual dashboard. This dashboard is fed into copilot and then the recommendation and suggestions are finally imported to Agentic AI and then generated into a chatbot – **'Campus Compass AI'**, presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

Describe the data cleaning process and present the before/after comparison of your dataset.

- Dataset source: The data was manually collected by generating questionnaire through google form to various faculty members.
- Original dataset size: [Number of rows and columns in the raw data.]
- Issues found: Some of the issues - duplicates, null values, formatting inconsistencies, recommendations and challenges not entered by few faculties.]
- Cleaning steps applied: [Describe each step performed in Excel / Power BI / Chatbot prompts.]
- Cleaned dataset size: 201 rows and 22 columns

Table – Summary of Faculty Dataset

Parameters	Individual Fields
Core Engagement KPI's	FDP/MDP Participation Rate, Average Teaching hours/week, Workload Balance Index
Satisfaction & Experience KPI's	Onboarding Satisfaction Score, Facilities Satisfaction Index , AI tools and digital support satisfaction
Career & Retention KPI's	Career Progression Clarity, Retention Intent , Stress Frequeuncy Index , Wellness Score , Net Promotor Score
Demographics	Age , Duration , Gender

S.No	Designation	Employment Duration	Gender	Age Group	FDP/MDP Attended in last year	Teaching Hours (weekly)	Workload balance perception	Onboarding Satisfaction Score	Net Promotor Score (NPS
1	Assistant Professor	Mid-Senior Faculty	Female	31-40	Between 5-10	40-50	Neutral (3)	Satisfied (4)	Neutral (3)
2	Professor	Junior Faculty	Male	Below 30	Between 0-5	30-40	Strongly Disagree (1)	Satisfied (4)	Neutral (3)
3	Associate Professor	Senior Faculty	Female	More than 50	More than 10	Less than 30	Disagree (2)	Neutral (3)	Unlikely (2)
4	Assistant Professor	Junior Faculty	Female	31-40	Between 5-10	40-50	Disagree (2)	Dissatisfied (2)	Very Likely (5)
5	Doctorate	Senior Faculty	Male	31-40	Between 0-5	30-40	Strongly Disagree (1)	Satisfied (4)	Likely (4)
6	Associate Professor	Mid-Senior Faculty	Female	41-50	Between 0-5	More than 50	Neutral (3)	Satisfied (4)	Unlikely (2)
7	Professor	Mid-Senior Faculty	Female	Below 30	Between 0-5	30-40	Disagree (2)	Dissatisfied (2)	Very Unlikely (1)
8	Assistant Professor	Mid-Senior Faculty	Male	31-40	Between 0-5	30-40	Strongly Disagree (1)	Neutral (3)	Neutral (3)
9	Professor	Mid-Senior Faculty	Female	41-50	More than 10	40-50	Disagree (2)	Highly dissatisfied (1)	Unlikely (2)
10	Assistant Professor	Mid-Senior Faculty	Male	41-50	Between 5-10	40-50	Disagree (2)	Highly dissatisfied (1)	Very Unlikely (1)
11	Assistant Professor	Mid-Senior Faculty	Male	Below 30	Between 0-5	30-40	Disagree (2)	Neutral (3)	Neutral (3)
12	Professor	Senior Faculty	Female	41-50	More than 10	Less than 30	Strongly Agree (5)	Satisfied (4)	Likely (4)
13	Associate Professor	Senior Faculty	Male	41-50	Between 5-10	More than 50	Disagree (2)	Dissatisfied (2)	Unlikely (2)
14	Professor	Senior Faculty	Female	41-50	Between 0-5	40-50	Disagree (2)	Dissatisfied (2)	Neutral (3)
15	Associate Professor	Senior Faculty	Female	31-40	Between 0-5	30-40	Disagree (2)	Dissatisfied (2)	Unlikely (2)
16	Assistant Professor	Senior Faculty	Female	More than 50	Between 0-5	40-50	Strongly Agree (5)	Dissatisfied (2)	Neutral (3)
17	Visiting Faculty	Mid-Senior Faculty	Female	31-40	More than 10	30-40	Strongly Disagree (1)	Neutral (3)	Unlikely (2)
18	Associate Professor	Junior Faculty	Female	41-50	Between 0-5	Less than 30	Neutral (3)	Satisfied (4)	Unlikely (2)
19	Associate Professor	Mid-Senior Faculty	Male	31-40	Between 0-5	More than 50	Neutral (3)	Neutral (3)	Neutral (3)
20	Professor	Senior Faculty	Female	41-50	More than 10	Less than 30	Strongly Disagree (1)	Neutral (3)	Neutral (3)
21	Assistant Professor	Junior Faculty	Male	More than 50	Between 5-10	30-40	Strongly Disagree (1)	Highly satisfied (5)	Very Unlikely (1)
22	Associate Professor	Senior Faculty	Female	41-50	Between 0-5	40-50	Strongly Agree (5)	Highly satisfied (5)	Likely (4)
23	Associate Professor	Mid-Senior Faculty	Female	41-50	More than 10	30-40	Strongly Agree (5)	Neutral (3)	Neutral (3)
24	Professor	Mid-Senior Faculty	Female	41-50	Between 5-10	More than 50	Disagree (2)	Highly dissatisfied (1)	Very Unlikely (1)

Table – Snapshot of Cleaned Faculty Data

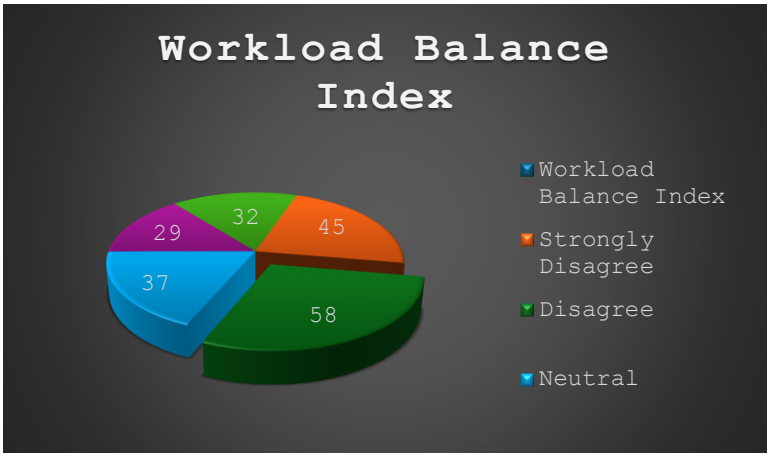
Career Progression Clarity	Retention Intent	Stress Frequency	Wellness Score	Facility Satisfaction Score	Research Grant Access	Technical Training	FDP/MDP Awareness	AI tools and digital support satisfaction	Interest Training
Neutral (3)	Very Likely	Mostly	3	Neutral (3)	No	Yes	Moderate	Satisfied	Yes
Not Clear (4)	Yes	Sometimes	4	Highly Satisfied (5)	No	No	Low	Dissatisfied	No
No opportunities (5)	Yes	Sometimes	1	Neutral (3)	Yes	No	Moderate	Very Dissatisfied	Yes
No opportunities (5)	Very Likely	Often	5	Neutral (3)	No	No	High	Neutral	Maybe
Not Clear (4)	Very Likely	Rarely	3	Highly Satisfied (5)	No	No	High	Neutral	Yes
Neutral (3)	Yes	Sometimes	3	Highly Dissatisfied (1)	No	Yes	Moderate	Satisfied	Maybe
Somewhat Clear (2)	Unlikely	Often	4	Highly Dissatisfied (1)	No	Yes	High	Satisfied	No
Not Clear (4)	Very Likely	Mostly	2	Neutral (3)	Yes	Yes	Low	Satisfied	Yes
Somewhat Clear (2)	Undecided	Often	4	Satisfied (4)	Yes	No	Moderate	Neutral	No
Very Clear (1)	Very Likely	Always	3	Highly Dissatisfied (1)	No	Yes	High	Very Satisfied	No
Somewhat Clear (2)	Very Likely	Sometimes	2	Highly Satisfied (5)	No	Yes	Moderate	Satisfied	No
Somewhat Clear (2)	Yes	Often	2	Neutral (3)	Yes	Yes	High	Very Satisfied	Maybe
Not Clear (4)	Very Likely	Often	1	Neutral (3)	No	No	High	Dissatisfied	No
Neutral (3)	Very Likely	Often	3	Neutral (3)	Yes	No	High	Dissatisfied	Yes
Neutral (3)	Yes	Always	5	Satisfied (4)	No	Yes	Moderate	Dissatisfied	Yes
Very Clear (1)	Yes	Rarely	3	Neutral (3)	No	Yes	High	Neutral	Maybe
Neutral (3)	Likely	Mostly	4	Neutral (3)	Yes	No	High	Very Dissatisfied	Yes
No opportunities (5)	Undecided	Rarely	5	Highly Dissatisfied (1)	No	Yes	Moderate	Satisfied	No
Neutral (3)	Yes	Always	1	Highly Dissatisfied (1)	No	Yes	Moderate	Very Satisfied	Yes
No opportunities (5)	Yes	Often	3	Satisfied (4)	No	Yes	Moderate	Satisfied	No
No opportunities (5)	Likely	Often	4	Highly Dissatisfied (1)	Yes	No	Low	Neutral	Maybe
Neutral (3)	Yes	Always	1	Neutral (3)	No	No	Moderate	Neutral	Yes
No opportunities (5)	Yes	Sometimes	5	Highly Dissatisfied (1)	No	No	High	Satisfied	Yes
Neutral (3)	Undecided	Sometimes	5	Satisfied (4)	Yes	No	Moderate	Dissatisfied	Maybe

Table – Snapshot of Cleaned Faculty Data

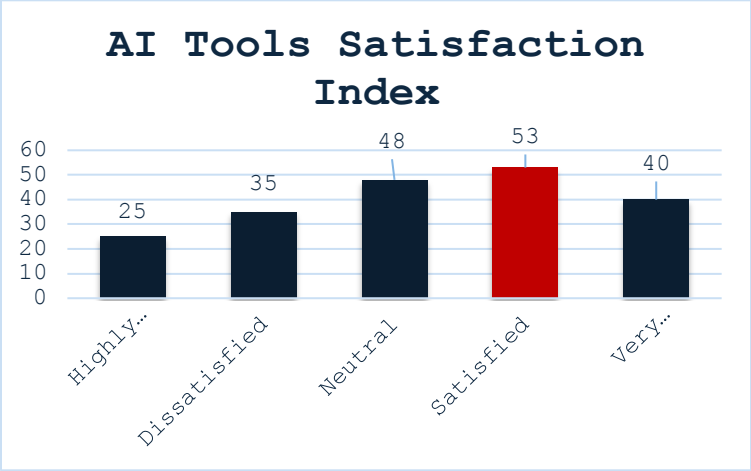
4.2 Visualization / Output Results

Present the key visualizations, dashboards, or generated content produced by your project.

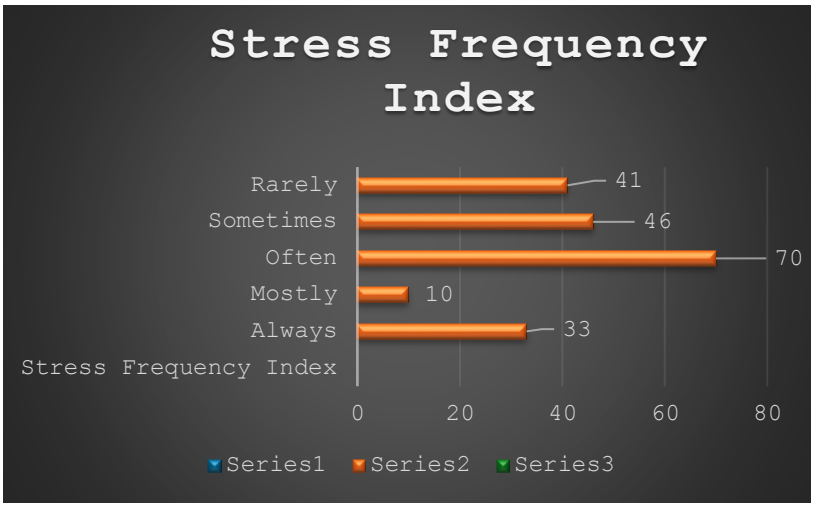
- **Dashboard / Chart 1:** As stated this pie chart shows the Workload Balance Index. As illustrated, about 58 out of 201 staffs disagreed to workload balance. Closely followed by 45 who strongly disagree.



- Dashboard / Chart 2: [Describe — AI tools satisfaction index] This shows how the HEI's support their staff in using AI based tools , techniques and introduce lectures or education.



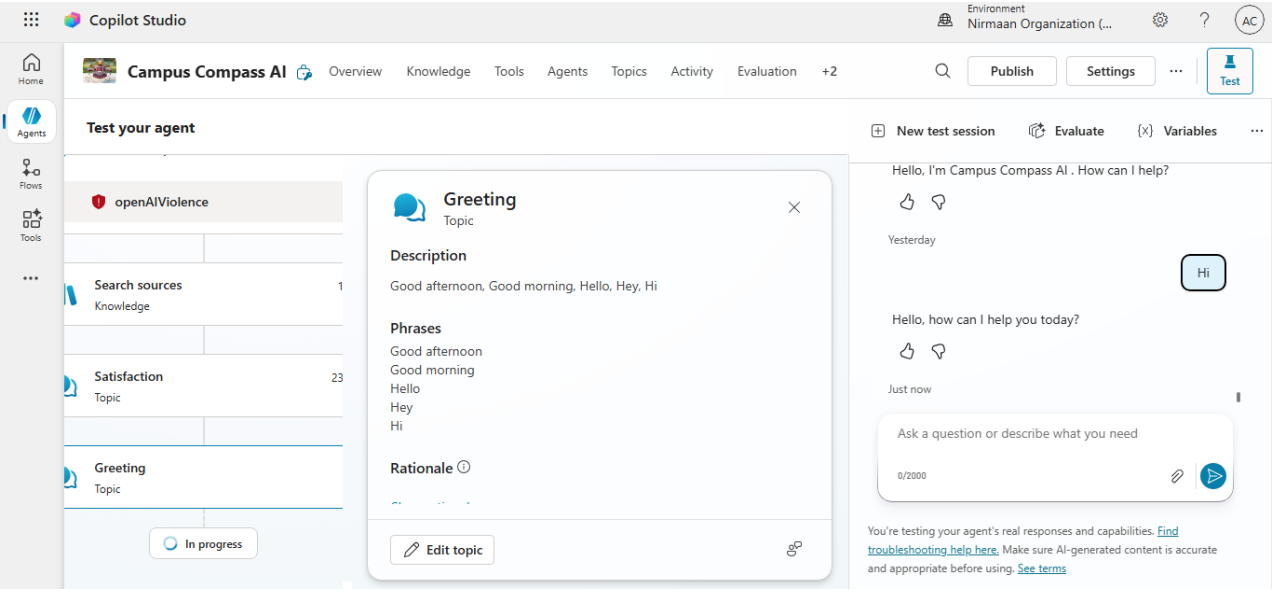
- Dashboard / Chart 3: [Stress Frequency Index] This shows the proportion of faculties dealing with workplace stress and anxiety.



4.3 AI Tool Integration Results

The final part of the project involves integrating the output received from visualization and AI generated inputs from Co-pilot and inputting the same to Microsoft Co-pilot Studio in the form of Agentic AI finally leading to the creation of the end product – **'Campus Compass AI – Smart Navigation Bot'**

- Tool used: [Microsoft Co-pilot / Google Gemini / Microsoft Co-pilot Studio]
- Prompts used: These prompts were used in the chatbot deployment
  a) *Be friendly, supportive, and professional in tone.*
  b) *Provide clear and concise answers.*
  c) *When possible, include links to official resources or documents.*



By integrating the suggestions and recommendations, finally an AI powered chatbot is created which helps in solving various queries of the faculty's real time and thereby locating resources, navigating the buildings etc. through a portable and conversation friendly device (it can also be downloaded and stored in the mobile phones)

CHAPTER 5
Discussion and Conclusion

5.1 Key Findings

Onboarding Satisfaction Score	
Highly Dissatisfied	30
Dissatisfied	53
Neutral	48
Satisfied	45
Highly Satisfied	24

FDP/MDP Participation Rate	
Between 0-5	125
Between 5-10	44
More than 10	32

Stress Frequency Index	
Always	33
Mostly	10
Often	70
Sometimes	46
Rarely	41

Facility Satisfaction Index	
Highly Dissatisfied	49
Dissatisfied	26
Neutral	54
Satisfied	48
Highly Satisfied	24

The following tables showcase the importance of creating a chatbot to check the underlying ambiguities and problems in the mind of the faculty members.

[**Key Finding 1:** After cleaning and summarizing the date, 26% of the faculties were dissatisfied with the on boarding which needs to be addressed be identifying their queries.

[**Key Finding 2:** 125 faculty members have individually participated in FDP/MDP programs showing an increases interest in upskilling.

[**Key Finding 3:** 34% of the faculties reported Workplace Stress this is quite alarming as if remains unchecked it can lead to underlying demotivation and reduced productivity levels and maybe attrition

[Key Finding 4: 5 faculties are neutral or a balanced viewpoint of the facilities provided like –labs, canteens, transportation, leaves, mediclaim. This means the faculty members are not clearly aware of the same.

All these findings strongly recommend the Agentic AI formation which can check the problems at their root and find meaningful solutions.

5.2 Project Links

Share the relevant links to your project work for evaluation.

- GitHub Link of the Project: https://github.com/Ashjosh-AI/Faculty-Onboarding-Bot-Project-Report---Campus-Compass-AI-

- Video Recording of Project Demonstration - file:///C:/Users/Ravi%20joshi/Documents/Zoom/2026-04-20%2020.57.48%20Ayesha%20Chakravarty's%20Zoom%20Meeting/video1755442249.mp4

5.3 Limitations

Every project has a few limitations hence, this project is itself not free from biases.

1. **Simulated Dataset** - Firstly the dataset is small and is a combination of real and AI based responses. This makes it not the best fit for deriving outputs. However, the responses are more or less helpful in generating an overview of the lacunae and coming with AI based solutions.
2. **Scalability Issues** – How feasible this chatbot will be for larger setups like – Tech parks and large hospitals needs to be tested before implementation.
3. **Dependence on Training Data** – AI chatbot is highly dependent on the knowledge base, hence if it has biases and the prompts are not engineered properly then it will be reflected in the responces.
4. **Integration with LMS and Maintenance -** It is not an easy task to integrate the chatbot with an institution LMS as well as to maintain the chatbot and run updates. Since technology is transforming at a rapid pace hence proper maintenance will need both investment and constanct monitoring to deliver perfect outputs.

5.4 Future Work

The **future of work** for faculty members using your ***Campus Compass AI*** chatbot looks promising because it shifts their daily experience from **manual, fragmented tasks** to **streamlined, intelligent support**.

- **Paperless Onboarding & Orientation** Faculty will no longer rely on lengthy paperwork; the bot can guide them through digital forms, policies, and institutional processes instantly.
- **Real-Time Administrative Support** Routine queries (leave policies, schedules, reporting formats) will be answered by the chatbot, reducing dependency on administrative staff and freeing time for teaching and research.
- **Personalized Faculty Assistance** The bot can act as a “personal assistant,” reminding faculty of deadlines, training opportunities, and FDP/MDP programs, tailored to their career stage.
- **Stress & Workload Reduction** By automating repetitive tasks and clarifying responsibilities, faculty can focus on high-value work like research, mentoring, and student engagement.
- **Continuous Learning & Development** Integrated with LMS, the chatbot can recommend AI workshops, technical training, and career progression resources, supporting lifelong learning.

5.5 Conclusion

In conclusion, the ***Campus Compass AI*** project demonstrates how combining structured data analysis with intelligent automation can transform the faculty experience in higher education institutions. By first identifying challenges through survey-based dashboards and then addressing them with **Campus Compass AI**, an LMS-integrated chatbot, the project reduces administrative burden, improves clarity in processes, and enhances faculty motivation and retention. This dual approach—diagnostics through Excel dashboards and real-time support through AI—offers a

sustainable pathway to increased productivity, streamlined on boarding, and a more supportive academic environment.

Although the project needs to be handled carefully at various levels and needs proper investment, time and maintenance but if integrated and running properly it can change the way work is done drastically and can be implemented to various other projects like – student query resolver, tourism bots, hospitals etc.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

[2] Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

[3] Microsoft Corporation. Microsoft Copilot — AI-Powered Productivity. Available at: https://copilot.microsoft.com

[4] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.

[5] (https://www.education.gov.in/sites/upload_files/mhrd/files/PIB2102789.pdf)

[6] (Jia, J. The Study of the Application of a Keywords-based Chatbot System on the Teaching of Foreign Languages. *arXiv* **2003**, arXiv:cs/0310018. [**Google Scholar**])

[7] (Cahn, J. CHATBOT: Architecture, Design, & Development. Ph.D. Thesis, University of Pennsylvania, School of Engineering and Applied Science, Philadelphia, PA, USA, 2017. [**Google Scholar**]).

[8] (Turing, A.M. Computing Machinery and Intelligence. *Mind* **1950**, *LIX*, 433–460. [**Google Scholar**])

[9] (A Literature Survey of Recent Advances in Chatbots by Guendalina Caldarini Kenneth McGarry)

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

S.No	Designation	Employment Durati	Gende	Age Group	FDP/MDP Attended in last yea	Teaching Hours (weekly)	Workload balance perceptio	Onboarding Satisfaction Scor	Net Promotor Score (NPS
1	Assistant Professor	Mid-Senior Faculty	Female	31-40	Between 5-10	40-50	Neutral (3)	Satisfied (4)	Neutral (3)
2	Professor	Junior Faculty	Male	Below 30	Between 0-5	30-40	Strongly Disagree (1)	Satisfied (4)	Neutral (3)
3	Associate Professor	Senior Faculty	Female	More than 50	More than 10	Less than 30	Disagree (2)	Neutral (3)	Unlikely (2)
4	Assistant Professor	Junior Faculty	Female	31-40	Between 5-10	40-50	Disagree (2)	Dissatisfied (2)	Very Likely (5)
5	Doctorate	Senior Faculty	Male	31-40	Between 0-5	30-40	Strongly Disagree (1)	Satisfied (4)	Likely (4)
6	Associate Professor	Mid-Senior Faculty	Female	41-50	Between 0-5	More than 50	Neutral (3)	Satisfied (4)	Unlikely (2)
7	Professor	Mid-Senior Faculty	Female	Below 30	Between 0-5	30-40	Disagree (2)	Dissatisfied (2)	Very Unlikely (1)
8	Assistant Professor	Mid-Senior Faculty	Male	31-40	Between 0-5	30-40	Strongly Disagree (1)	Neutral (3)	Neutral (3)
9	Professor	Mid-Senior Faculty	Female	41-50	More than 10	40-50	Disagree (2)	Highly dissatisfied (1)	Unlikely (2)
10	Assistant Professor	Mid-Senior Faculty	Male	41-50	Between 5-10	40-50	Disagree (2)	Highly dissatisfied (1)	Very Unlikely (1)
11	Assistant Professor	Mid-Senior Faculty	Male	Below 30	Between 0-5	30-40	Disagree (2)	Neutral (3)	Neutral (3)
12	Professor	Senior Faculty	Female	41-50	More than 10	Less than 30	Strongly Agree (5)	Satisfied (4)	Likely (4)
13	Associate Professor	Senior Faculty	Male	41-50	Between 5-10	More than 50	Disagree (2)	Dissatisfied (2)	Unlikely (2)
14	Professor	Senior Faculty	Female	41-50	Between 0-5	40-50	Disagree (2)	Dissatisfied (2)	Neutral (3)
15	Associate Professor	Senior Faculty	Female	31-40	Between 0-5	30-40	Disagree (2)	Dissatisfied (2)	Unlikely (2)
16	Assistant Professor	Senior Faculty	Female	More than 50	Between 0-5	40-50	Strongly Agree (5)	Dissatisfied (2)	Neutral (3)
17	Visiting Faculty	Mid-Senior Faculty	Female	31-40	More than 10	30-40	Strongly Disagree (1)	Neutral (3)	Unlikely (2)
18	Associate Professor	Junior Faculty	Female	41-50	Between 0-5	Less than 30	Neutral (3)	Satisfied (4)	Unlikely (2)
19	Associate Professor	Mid-Senior Faculty	Male	31-40	Between 0-5	More than 50	Neutral (3)	Neutral (3)	Neutral (3)
20	Professor	Senior Faculty	Female	41-50	More than 10	Less than 30	Strongly Disagree (1)	Neutral (3)	Neutral (3)
21	Assistant Professor	Junior Faculty	Male	More than 50	Between 5-10	30-40	Strongly Disagree (1)	Highly satisfied (5)	Very Unlikely (1)
22	Associate Professor	Senior Faculty	Female	41-50	Between 0-5	40-50	Strongly Agree (5)	Highly satisfied (5)	Likely (4)
23	Associate Professor	Mid-Senior Faculty	Female	41-50	More than 10	30-40	Strongly Agree (5)	Neutral (3)	Neutral (3)
24	Professor	Mid-Senior Faculty	Female	41-50	Between 5-10	More than 50	Disagree (2)	Highly dissatisfied (1)	Very Unlikely (1)

Indicated a sample cleaned dataset showing 24 entries of the 201 entries.

Appendix B — Key Excel Formulas / DAX Measures Used

List any important formulas, functions, or DAX measures used in the project:

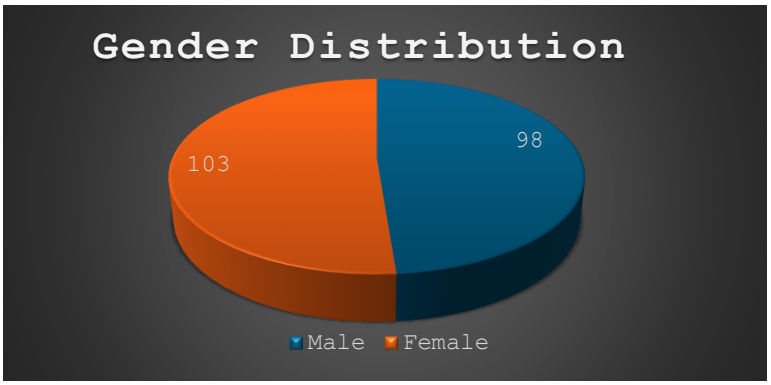
- =COUNTIF(range, criteria) — used to count duplicates
- =SUMIF ((range, criteria) – used to do a summation

Appendix C — AI Prompts Used

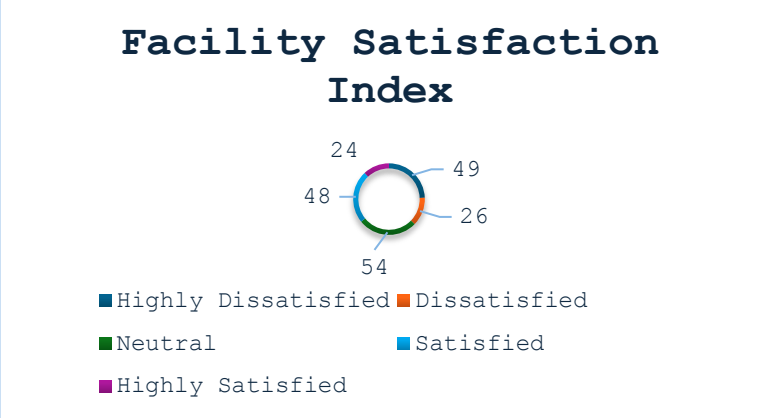
List the key prompts you used with Copilot or AI Chatbot during the project:

- Prompt 1: "Summarize the dashboard and identify challengges."
- Prompt 2: "Provide a DFC Level) diagram”
- Prompt 3: "Provide an icon for the agentic AI Chabot- Campus Compass AI"
- Prompt 4 : " Generate a small paragraph for conclusion of the project”

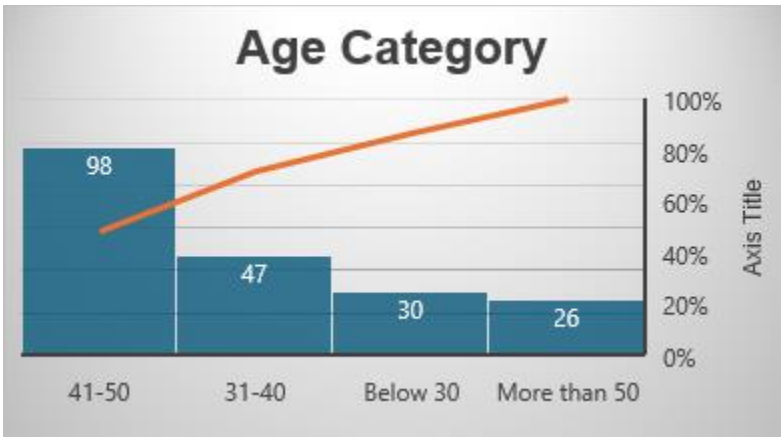
Appendix D — Screenshot Gallery



This shows the gender distribution of faculties. 103 are females and 98 are males, indicating an almost balanced gender in thr workforce.



This image shows the facility satisfaction among faculties. Most of them (54) are neutral followed by highly dissatisfied (49) which is a red flag.



This chart captures the age distribution. It indicates that almost 80% of the faculties are between 40's-50's showing a mature and senior faculty members.